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Prediction of 1-year mortality after heart transplantation using machine learning approaches: A single-center study from China

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ABSTRACT

Background: Heart transplantation (HTx) remains the gold-standard treatment for end-stage heart failure. The aim of this study was to establish a risk-prediction model for assessing prognosis of HTx using machine-learning approach.

Methods: Consecutive recipients of orthotopic HTx at our institute between January 1st, 2015 and December 31st, 2018 were included in this study. The primary outcome was 1-year mortality. Least absolute shrinkage and selection operator method was used to select variables and seven different machine-learning approaches were employed to develop the risk-prediction model. Bootstrap method was used for model validation. Shapley Additive exPlanations (SHAP) method was used for model interpretation.

Results: 381 recipients were included with average age of 43.783 years old. Albumin, recipient age and left atrium diameter ranked top three most important variables that affected the 1-year mortality of HTx. Other important variables included red blood cell, hemoglobin, lymphocyte%, smoking history, use of lyophilized rhBNP, use of Levosimendan, hypertension, cardiac surgery history, malignancy and endotracheal intubation history. Random Forest (RF) model achieved the best area under curves (AUC) of 0.801 and gradient boosting machine (GBM) showed the best sensitivity of 0.271. SHAP method was introduced to display the RF model's predicting processes of "survival" or "death" in individual level.

Conclusions: We established the risk-prediction model for postoperative prognosis of HTx patients by using machine learning method and demonstrated that the RF model performed the highest discrimination with the largest AUC when validated. This prediction model could help to recognize high-risk HTx recipients, provide personalized therapy plan and reduce organ wastage.

1. Introduction

Heart transplantation (HTx) remains the gold-standard treatment for patients with end-stage heart failure refractory to medical treatment and/or intervention procedure [1,2]. However, due to the shortage of donor organs and numerous factors which complicate searching, acquiring and successfully transplanting a well-matched heart, the use of transplantation is severely restricted. Decision regarding transplant candidacy and donor organ allocation are influenced by expectation of posttransplant survival, which has placed statistical prediction of recipient's survival at the forefront of the debate. Multivariable regression models are the most widely used model for survival estimation at present. Nonetheless, the complexity of organ transplantation and the wide range of possible serious complications resulted in limited prediction accuracy.

Machine learning techniques are emerging as a rapid and powerful way to understand intricate patterns from massive, complex and heterogeneous data. In addition, machine learning could develop standardized predictive models which would help clinicians in making patient-specific decisions [3]. Modern machine learning algorithms

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such as random forest (RF) and support vector machines (SVM) have been widely used in clinical data mining with satisfactory performance. Compared with traditional statistical models, machine learning models usually yield lower prediction error. Typically, data are divided into training and test data sets. The algorithm trains the model based on the training set and learn to predict the outcomes using the observable variables, which were called "features", from testing data set. The performance of the learned model was quantified by measuring the difference between the held-out testing data and the predicted ones [4].

To date, there is no generally accepted and high-accuracy risk-prediction model for assessing prognosis after HTx. Besides, current studies deliberated to develop mortality prediction models are mostly based on United Network for Organ Sharing (UNOS) registry in which Chinese heart transplant patients were not included. In the present study, seven machine learning approaches were applied to predict the overall 1-year survival of HTx patients and Shapley Additive exPlanations (SHAP) [5] method was used to visualize the process of how a machine learning model came to a decision. This survival prediction model may help the clinical decision-making process, optimize existing organ allocation strategies and guide routine management of recipients in the HTx waitlist.

2. Methods

2.1. Ethical statement

All donor organs were donated to the Red Cross Society and allocated by the China Organ Transplant Response System based on Chinese laws. Formal approval was obtained through the ethics committee of Tongji Medical College of Huazhong University of science and Technology (IORG No.: IORG0003571). The study was in line with the national program for deceased organ donation in China (national protocol for China category I). The study protocol also met the ethical standards of the Declaration of Helsinki and Istanbul [6].

2.2. Study population and data elements

Consecutive recipients of orthotopic heart transplantation at our institute between January 1st, 2015 and December 31st, 2018 were included in this study. The primary outcome was postoperative 1-year mortality, including in-hospital mortality and 1-year follow-up mortality. Patients with survival time less than one year and patients undergoing re-transplant or multiple organ transplant were excluded. Data for each patient were obtained retrospectively from the electronic medical records and subsequently through outpatient review and telephone. Follow-up ended on February 26th, 2020. We discarded variables with missing proportion more than 10%. The remaining 70 variables were imputed appropriately. To be specific, continuous variables were imputed with column mean and categorical variables were imputed with most frequent values in the corresponding column.

2.3. Statistical analysis

2.3.1. Exploratory analysis

The original data set was randomly split into a 70% training set and a 30% testing set. On the development set, descriptive measures were applied to identify the potential variables to be selected in the model. Univariate analysis for continuous variables were conducted by independent sample *t*-test if values were normally distributed. Mann-Whitney *U* test was conducted if normality assumption were violated. Categorical variables were tested by chi-squared test or Fisher's exact test if expected cell count less than five. Variables with *p* values less than 0.1 in the development set were considered having decent significance impact on death of patients in one year after surgery, thus should be considered in variable selection procedure. All univariate analysis were performed by Python (Version 3.7, Anaconda Distribution, https://www.anaconda.com).

2.3.2. Model development and evaluation

Least absolute shrinkage and selection operator (LASSO) method [7] were used to further select variables from those with *p*-value less than 0.1 in the univariate analyses in order to alleviate possible overfitting, thus improve model performance. LASSO variable selection approach was implemented by Sklean package (Version 0.23.0, https://scikit-learn.org/stable/) with Python. Seven different machine learning approaches including logistic regression (LR), SVM, RF, extreme gradient boosting (XGBoost), adaptive boosting (AdaBoost), gradient boosting machine (GBM) and artificial neural network (ANN) were employed. Since the choice of tuning parameters is essential for model performance, we used grid search method with five-fold cross validation to choose parameters for each of the models. XGBoost model was implemented using xgboost package (https://xgboost.readthedocs.io/en/latest/python/index.html). LR, SVM, RF, AdaBoost and GBM model were implemented using Scikit-learn package (Version 0.23.0, https://scikit-learn.org/stable/). ANN model was implemented using Keras deep learning framework (https://keras.io/) with TensorFlow (https://www.tensorflow.org/) backend.

We used bootstrap approach to evaluate the models' performance on the validation set. Bootstrap samples were constructed by extracting samples from original data set with replacement with the same sample size as validation set [8]. The bootstrap procedure was repeated 1000 times and 95% confidence interval were calculated to evaluate model stability. Main model evaluation metric included area under receiver-operating curve (AUC), calculated as the area under receiver operating curve. Model accuracy, sensitivity and specificity were also calculated. For model interpretation, the Shapley Additive exPlanations was implemented using Python shap package (https://shap.readthedocs.io/en/latest/).

3. Results

3.1. Baseline characteristics

Fig. 1 showed the whole workflow of the present study. A total of 381 recipients were included in the present study with average age of 43.783 ± 16.453 years old. Male accounted for 76.2% (273 patients). The overall 1-year survival rate of this cohort was 84.8%. The etiology of heart transplantation included cardiomyopathy, coronary artery disease, valvular heart disease, congenital heart disease and so on, of which 214 were dilated cardiomyopathy (65.2%). The data distribution in the training and testing set were rather uniform, in which only the *p* value of recipient's use of adrenalin hydrochloride and donor's weight were less than 0.05. The detailed preoperative baseline characteristics of both HTx recipients and donors were listed in Table 1 and Supplementary Table 1.

3.2. Variables selection

There were 13 variables with *p*-value less than 0.1 in the training set including albumin (ALB), recipient age, diameter of left atrium (LA) from ultrasonic cardiography, red blood cell count (RBC), hemoglobin (Hb), lymphocyte proportion (L%), smoking history, use of lyophilized recombinant human brain natriuretic peptide (rhBNP), use of Levosimendan, hypertension, cardiac surgery history, malignancy and endotracheal intubation history. The threshold 0.1 was chosen to encourage more promising features to enter the following model. A LASSO feature selection approach tuned by five-fold cross validation was applied on the 13 features. However, no feature was eliminated, which indicating all features showed nonnegligible effect in affecting model performance (Supplementary Fig. 1).

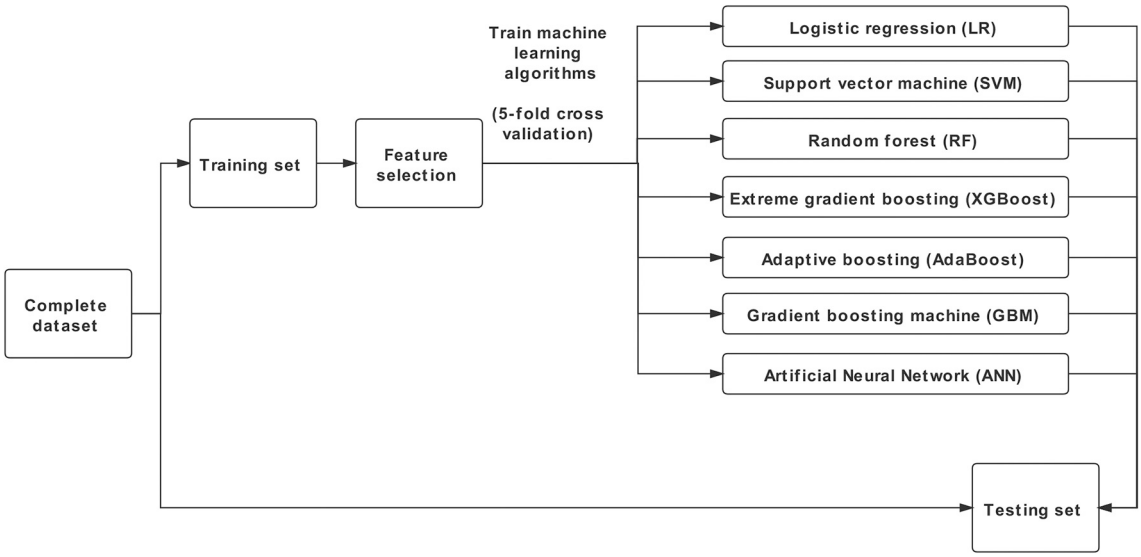

Fig. 1. Workflow diagram: the complete data set were randomly split into 70% training set and 30% testing set. Feature selection was applied on the training set. Seven different machine learning model were trained and tuned using 5-fold cross validation. Lastly, trained models were evaluated on the testing set.

3.3. Model performance and interpretability

The involved candidate variables and major evaluation metrics of the testing set using bootstrap validation in predicting 1-year survival were summarized in Table 2. Receiver operation curves (ROC) of 7 different models were showed in Fig. 2. RF model achieved the best performance among all seven machine learning models, reaching the best AUC of 0.801(95% CI: 0.697–0.891). GBM model showed slightly lower AUC of 0.786(95% CI: 0.661–0.896). SVM model failed to discover any positive samples in test data set with sensitivity equaled to zero. In general, sensitivities were poor in all the models. GBM model showed the highest sensitivity of 0.271(95% CI: 0.077–0.533), which means GBM model could discover 27.1% of positive patients in the test data set. The sensitivity of RF model was 0.268(95% CI: 0.059–0.529). These results indicated that RF and GBM may be the best machine learning models to predict one-year mortality of HTx patients. Since the difference between RF and GBM was minor, RF model was taken as an example for model interpretability.

The feature importance ranking of RF model based on mean decrease of impurity was shown in Fig. 3. Higher the mean decrease of impurity means greater impact of the corresponding variable to prediction result. ALB, RBC, recipient age, LA and L% showed top-five significant discriminative contributions to RF model. Shapley values interpretation method was applied on RF model to see feature contribution to model output on individual level. Feature importance plot based on Shapley-values was displayed in Fig. 4, which showed major consistency with feature importance of RF model based on mean decrease of impurity with same top three features. Considering the top three features, a higher ALB value makes the patients less likely to die, on the contrary, enlarged LA or older recipient age makes the patients suffer more risk of death within one year after surgery.

Fig. 5 showed how these predictions were made by RF model of two individual patients in testing dataset. Patient 1 was a 20-year-old male diagnosed with dilated cardiomyopathy. He was in NYHA class IV with preoperative LVEF of only 14.9% and six-minute walking test (6MWT) of 280 m. The liver and renal function was approximately normal. The predicted probability for mortality was rather low for this patient at 25% compared with the average mortality of the test cohort (35%). The variables figured out by the model for predicting lower mortality included younger recipient age (20 y), small LA diameter (4.1 cm) and sufficient albumin (44.6 g/l). Although the risk was slightly elevated by the variables like RBC count (3.35 × 10^9/l) and L% (6.1%), the RF model still predicted outcome as survival for this patient, which was consistent with the actual outcome (true negative).

Patient 2 was a 24-year-old male diagnosed with complex congenital heart disease including single ventricular, tricuspid valve atresia, severe mitral regurgitation and pulmonary stenosis, complete transposition of great arteries and dilatation of the aortic sinuses and the ascending aorta. He couldn't tolerate 6MWT, the preoperative NT-pro-BNP was 11,665 pg/ml and LVEF was 16%. The liver and renal function was slightly abnormal with total bilirubin (TB) 59.5 μmol/l, serum creatinine (sCr) 133.5 μg/ml and blood urea nitrogen (BUN) 24.67 mg/dl. The patient used endotracheal intubation due to the extremely severe mixed pulmonary ventilation dysfunction. He had the mortality risk of 50% predicted by the model with low ALB of 32.4 g/l, enlarged LA diameter of 9.1 cm, use of Levosimendan and history of endotracheal intubation before surgery and so on. Recipient age (24 y) was the only variable that partially offset the high risk of mortality. This was a true-positive case with the outcomes of both model prediction and the true picture were death.

4. Discussion

With the rapid development of HTx in China, the imbalance between the organ demand and donation is gradually emerging. A feasible prognosis prediction model would help clinicians distinguish high-risk patients and optimize organ allocation strategies, making patients benefit most from HTx. Herein, for the first time, the present study assessed the predicting performance of different machine learning algorithms for overall 1-year mortality after HTx in China. The major novel findings are as follows: (I) RF model performed optimal predictive power with the best AUC of 0.801 and GBM model owned slightly lower AUC of 0.786. However, sensitivity was rather low for all the models with GBM achieving the best sensitivity of 0.271; (II) Top-five most importance variables related to the short-term prognosis of HTx included ALB, age, LA, RBC and Hb level of recipient; (III) Shapley-value shed light into model explanation among variables and displayed the prediction process in individual level. To our knowledge, our study presented the first 1-year survival prediction model of HTx for Chinese population using machine learning approach, and introduced Shapley-value to the field of HTx for the first time to interpretate and present the process of prediction.

Of all the establishing approaches, machine learning has been an up-and-coming method as they showed more advantages in finding

Table 1
Main baseline characteristics of HTx recipients. Continuous variables were shown by mean with standard deviation in bracket. Categorical variables were shown by frequency and corresponding proportion in each group.

	All	Training set	Testing set	*p value*
Age, years	43.783 (16.453)	44.211 (16.183)	42.798 (17.101)	0.580
Gender, male	249(0.759)	175(0.764)	74(0.747)	0.803
Weight, kg	62.516 (16.544)	63.098 (16.551)	61.131 (16.535)	0.467
Dilated cardiomyopathy	214(0.652)	151(0.659)	63(0.636)	0.744
Blood type				
A	109(0.332)	80(0.349)	29(0.293)	0.385
B	86(0.262)	57(0.249)	29(0.293)	0.487
AB	28(0.085)	18(0.079)	10(0.101)	0.652
O	97(0.296)	67(0.293)	30(0.303)	0.953
Past history				
Cardiac surgery history	88(0.268)	62(0.271)	26(0.263)	0.969
ECMO history	1(0.003)	0(0.0)	1(0.01)	0.670
IABP history	4(0.012)	2(0.009)	2(0.02)	0.756
Endotracheal intubation history	6(0.018)	3(0.013)	3(0.03)	0.545
CPR history	19(0.058)	11(0.048)	8(0.081)	0.336
Family history of cardiovascular disease	10(0.03)	8(0.035)	2(0.02)	0.708
Hypertension	50(0.152)	33(0.144)	17(0.172)	0.649
Hyperlipidemia	12(0.037)	8(0.035)	4(0.04)	0.932
Diabetes mellitus	41(0.125)	28(0.122)	13(0.131)	0.975
Chronic lung disease	9(0.027)	5(0.022)	4(0.04)	0.568
Chronic liver disease	28(0.085)	18(0.079)	10(0.101)	0.66
Chronic renal disease	17(0.052)	10(0.044)	7(0.071)	0.463
Stroke	8(0.024)	6(0.026)	2(0.02)	0.952
Malignancy	5(0.015)	5(0.022)	0(0.0)	0.320
Smoking history	116(0.354)	88(0.384)	28(0.283)	0.096
Donor information				
Donor gender, male	281(0.857)	196(0.856)	85(0.859)	0.805
Donor age, years	32.527 (12.7)	33.153 (12.632)	31.081 (12.804)	0.181
Donor weight, kg	61.677 (12.445)	62.351 (12.293)	60.126 (12.717)	0.033*
Cause of death				
Brain injury	196(0.598)	139(0.607)	57(0.576)	0.684
Cerebral hemorrhage	90(0.274)	61(0.266)	29(0.293)	0.719
Brain tumor	16(0.049)	10(0.044)	6(0.061)	0.708
Others	25(0.076)	18(0.079)	7(0.071)	0.983
Cold ischemic time, min	336.32 (111.843)	340.633 (110.844)	326.343 (114.063)	0.125

inherent correlations of complex and high-dimension data. Machine learning has been widely used in the field of clinical diagnosis and prognosis prediction. Compared with traditional regression analysis, machine learning methods, especially ensemble-based RF and GBM are more resist to overfitting [9,10]. Only a few studies focusing on predicting prognosis of HTx using machine-learning approaches have been published in the literature yet. Medved et al. [11] applied machine learning to compare the predictive performance of the existing International Heart Transplantation Survival Algorithm (IHTSA) and Index for Mortality Prediction After Cardiac Transplantation (IMPACT) for 1-year survival after HTx, and found that the IHTSA model (AUC = 0.654) showed superior power than the IMPACT model (AUC = 0.608). Miller and colleagues [12] used RF, ANN and classification and regression trees (CARF) to predict 1-, 3- and 5-year mortality after pediatric HTx and found RF performed best with AUC of 0.72 for 1-year mortality. However, sensitivity was poor across these models. Based on the

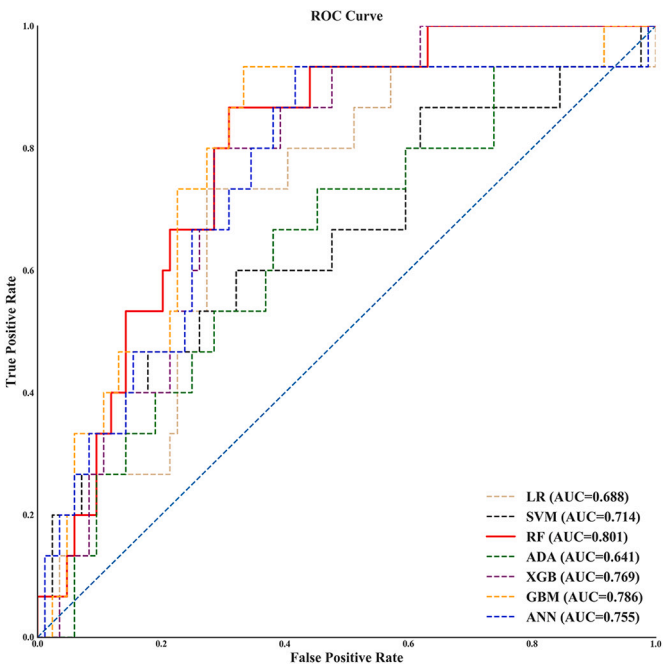

Fig. 2. Receiver operating curves of all seven machine learning models in the bootstrap validation. A greater area under receiver operating characteristic curve (AUC) represents better discriminative ability of the model.

Table 2
Model performance under bootstrap validation.

	Accuracy	AUC	Sensitivity	Specificity	Candidate variables
RF	0.828 (0.747–0.899)	0.801 (0.697–0.891)	0.268 (0.059–0.529)	0.927 (0.866–0.976)	ALB, recipient age,
AdaBoost	0.798 (0.717–0.869)	0.641 (0.479–0.788)	0.260 (0.059–0.500)	0.894 (0.821–0.954)	LA diameter, RBC,
LR	0.807 (0.727–0.879)	0.688 (0.549–0.816)	0.201 (0.000–0.429)	0.917 (0.851–0.966)	Hb, L%,
SVM	0.849 (0.778–0.919)	0.714 (0.574–0.834)	0.000 (0.000–0.000)	1.000 (1.000–1.000)	smoking history, lyophilized rhBNP,
XGBoost	0.828 (0.747–0.899)	0.769 (0.662–0.869)	0.138 (0.000–0.353)	0.953 (0.902–0.989)	Levosimendan, hypertension,
GBM	0.819 (0.737–0.889)	0.786 (0.661–0.896)	0.271 (0.077–0.533)	0.916 (0.845–0.966)	cardiac surgery history, malignancy,
ANN	0.849 (0.778–0.919)	0.755 (0.639–0.851)	0.066 (0.000–0.214)	0.988 (0.962–1.000)	endotracheal intubation history
Naïve	0.848 (0.778–0.909)	0.500 (0.500–0.500)	0.000 (0.000–0.000)	1.000 (1.000–1.000)	

95% confidence intervals were shown in parentheses. RF: random forest; AdaBoost: adaptive boosting; LR: linear regression; SVM: support vector machines; XGBoost: extreme gradient boosting; GBM: gradient boosting machine; ANN: artificial neural network; Naïve: naïve classifier that predict all samples with negative class.

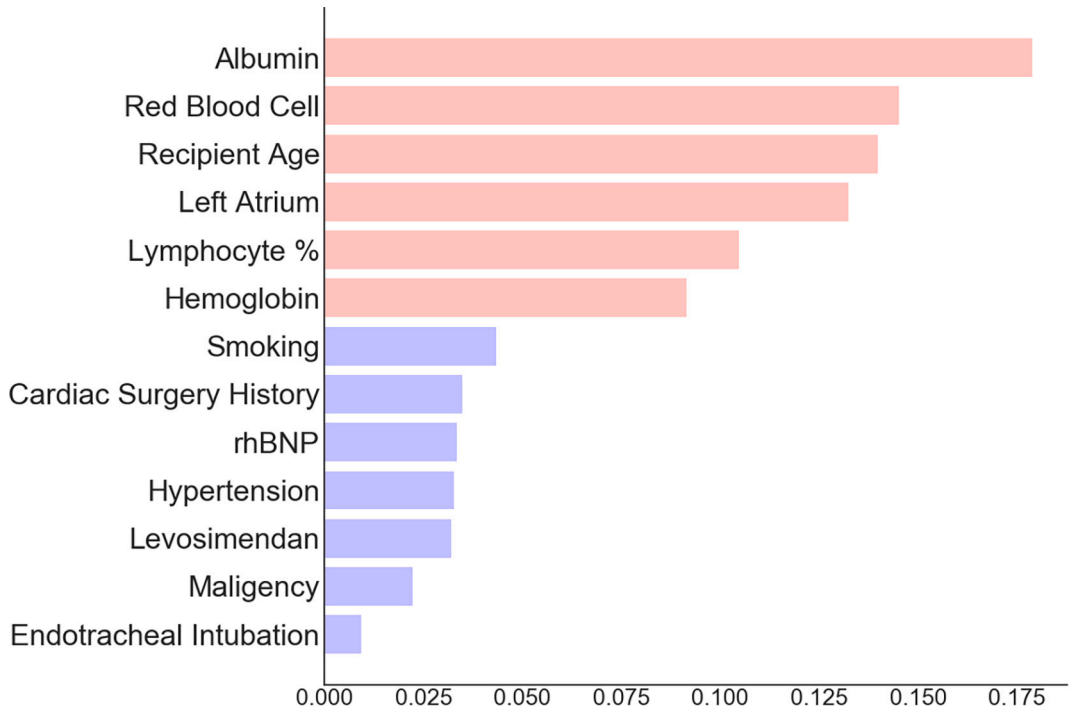

Fig. 3. Feature importance illustrated by Random Forest model. Variable importance was calculated by the sum of the decrease in error when split by a variable. The relative importance was the variable importance divided by the highest variable importance value such that values were bounded between 0 and 1. Variables showing in pink indicated higher importance. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Fig. 4. Feature importance based on Shapley Values from top down. Higher feature values were shown in red and lower feature values shown in blue. Albumin, left atrium diameter and recipient age were ranking top three most important features in affecting model output. In addition, most features showed discriminative ability to the model (red and blue dots were divided by the central vertical line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

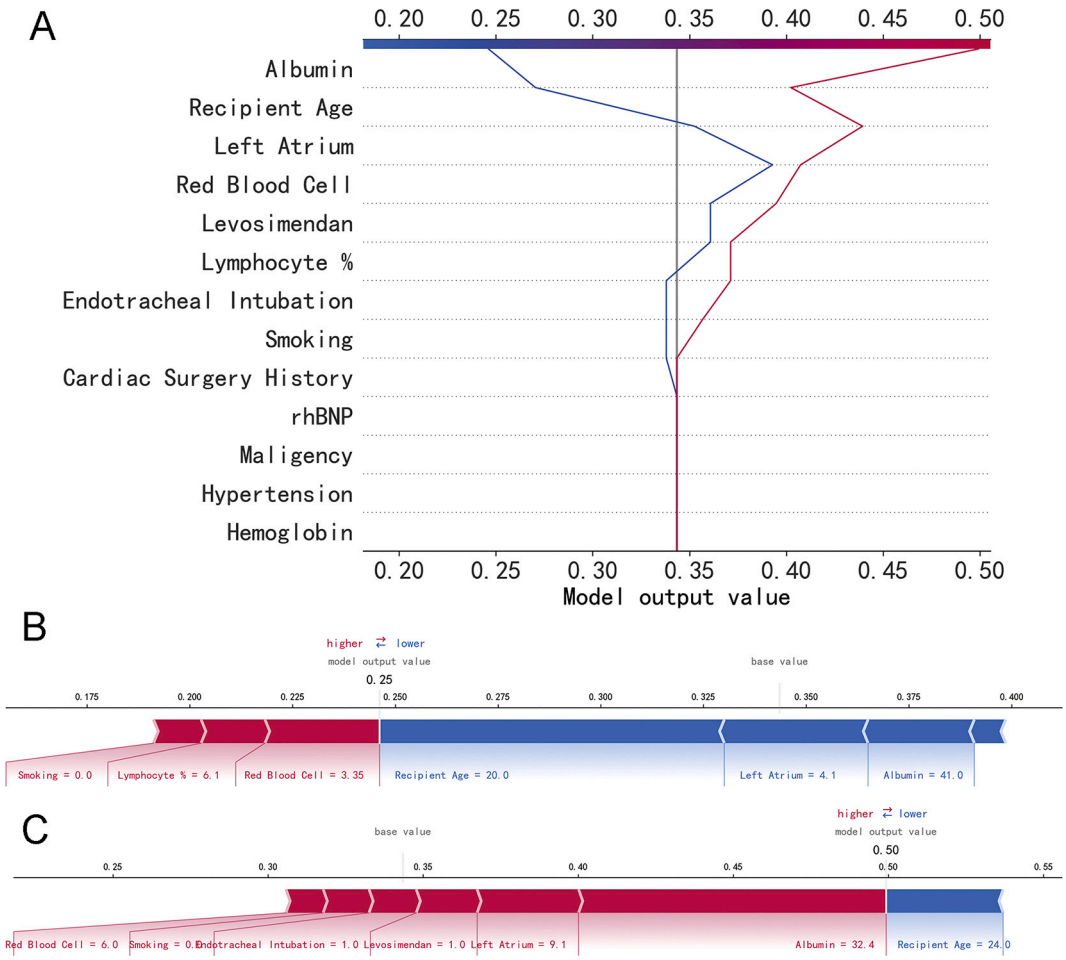

Fig. 5. A: SHAP decision plot for two patients in the testing set, one was predicted as "positive" and the other was predicted as "negative"; B: SHAP force plot for patient 1(survival); C: SHAP force plot for patient 2(death).

International Society of Heart and Lung Transplant (ISHLT) registry data, Agasthi et al. [13] developed a risk prediction model with the help of a GBM model. The model performed well in predicting 5-year mortality and graft failure, with AUC reaching 0.717 and 0.716, respectively. Significant variables with strong correlation with mortality were identified including recipient and donor age and BMI, bilirubin, renal function, history of mechanical ventilation, dialysis, diabetes and so on [13,14].

In the present study, we compared seven commonly used machine learning approaches in predicting 1-year mortality after HTx. In general, as an unbalanced task, most machine learning algorithms tend to perform like a naïve guess. For example, SVM model predict all cases in test data negative and did not make any positive prediction. In this situation, SVM showed no prediction ability and hence should not be applied for HTx patients. RF model outperformed other models in the study with accuracy of 0.828, AUC of 0.801, sensitivity of 0.268 and specificity of 0.927. In this case, the RF model reached decent performance in the external validation while the AUC of other reported models [12,15] were mostly between 0.6 and 0.7. On the other hand, however, as short time death is one of the most primary negative outcomes after HTx which we should prevent in any cases, a model that can make more accurate positive predictions would be preferable. The sensitivity of RF model was 0.268, indicating that only 26.8% of positive patients in the test data set would be found by RF model, which is barely satisfactory.

As the development of electronic medical record system, the application of machine-learning methods for prognosis-predicting and decision-making is getting extended. However, many models like XGBoost are the black box models which we cannot know the exact contribution of each feature to the decision for a specific case. Interpretability of model can be defined as the extent to which a human can understand the cause of the machine learning model's prediction [16]. The higher the interpretability of the model, the easier it will be for clinicians to understand the model's behavior and trust the model's conclusion, so as to make appropriate clinical decisions in the best interests of the patient [17]. With regard to this, we introduced the Shapley-value to provide explanations for the RF model. The SHAP method is a Game Theory-based method, within which the individual features act as players in a prediction task and the Shapley-value helps us fairly distribute the prediction performance among the features [5]. SHAP method possesses three main properties [17–19]: (1) Local Accuracy, (2) Missingness, (3) Consistency. Moreover, SHAP method are better aligned with human intuition and could support explanation of multiple model type including tree-based models, neural network and ensemble models.

By using SHAP method, we could shine light into the black box of machine learning models. According to the feature importance based on Shapley-values, higher value of ALB and complete blood counts (CBC) reduced the risk of early mortality after HTx, whereas older recipient age and larger LA diameter, use of Levosimendan and lyophilized rhBNP, hypertension, cardiac surgery history and endotracheal intubation increased the risk of death. Younger age and higher ALB and CBC value reflected better preoperative general conditions and nutritional status, while hypertension, surgery and intubation showed poor physical condition and increased the complexity of surgical process. Enlarged left atrium and using higher dose of Levosimendan and lyophilized rhBNP before operation implied more severe cardiac dysfunction, and thus

related to higher risk of death. It must be stated that smoking history was unexpectedly figured out as a protective variable for mortality, which was inconsistent with common sense and other researches [20,21]. In fact, our previous study [22] also found smoking history as a protective variable using Cox proportional hazard model. We think this surprising result may be due to the characteristics of our sample and need further verified by large-sample, multi-center studies.

Collectively, the present study proposed the first short-term prognosis prediction model for HTx based on Chinese population using machine learning approach. We demonstrated that the RF model performed the highest discrimination with the largest AUC when validated. Albumin, diameter of left atrium and recipient age ranked the top three most significant features that exert influence to the 1-year survival of HTx. Under the combined influence of these variables, an individual was predicted as "survival" or "death". As shown in the Fig. 5, with the help of Shapley-value, a doctor could easily and intuitively understand how the machine learning model makes the decision, and consequently trust and apply the model for clinical decision-making. This prediction model may help to recognize high-risk HTx recipients, optimize preoperative management in clinical practice and reduce organ wastage. It should be noted that the precision of positive predictions has yet to be improved. Further study is warranted in multicenter with a large sample to establish a HTx prognosis prediction model with more sufficient validity and reliability.

4.1. Limitation

This study was not devoid of limitations. Compared with traditional prediction model, machine learning method is particularly good at efficiently processing of "big data" sources, exploiting the complex, nonlinear relationships in a comprehensive and unbiased fashion [23]. Therefore, it is conceivable that the model performance largely depends on sample size of the original data. The sample size in the current study was relatively small from a machine learning perspective, especially the number of positive events (1-year mortality in this case), and might be partially responsible for the poor sensitivity of the prediction model. Furthermore, the retrospective clinical dataset was restricted by accuracy of data entry and variables collected, and potential recall bias and data incompleteness for a significant proportion of the patients would inevitably exist. This might blunt phenotyping of complex patients and attenuate the predictive ability of machine learning techniques.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijcard.2021.07.024.

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Declaration of Competing Interest

The authors declare no conflicts of interest.

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